MAPS OUT OF AN ALMOST FREE MODULE TO A BOUNDED BELOW COMPLEX IN THE DERIVED CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The two cross-chapter references inside the counted proof span (derived-lemma-morphisms-from-projective-complex and derived-lemma-morphisms-into-injective-complex) are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Allowed standard prerequisites (editorial).

- Almost free modules and their filtrations by free modules. fixed earlier in the same section and used throughout the counted spans. Supplies the filtration along which the lift is constructed.
- Bounded below complexes and the standard representatives of their derived objects. the bounded below hypothesis in the counted statement span, and the two cited derived-category lemmas in the counted proof span. Is what makes the successive obstructions computed in the lift vanish.

Source reference. Original lemma 064B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7860–7860. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 05TG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7480–7480. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let R be a ring. Let $M^\bullet$ be a complex of modules over R with $M^i = 0$ for $i > 0$ and M^0 a projective R-module. Let $K^\bullet$ be a second complex.*

1. *Assume $K^i = 0$ for $i \leq -2$. Then $\mathrm{Hom}_{D(R)}(M^\bullet, K^\bullet) = \mathrm{Hom}_{K(R)}(M^\bullet, K^\bullet)$.*
2. *Assume $K^i = 0$ for $i \notin [-1, 0]$ and K^0 a projective R-module. Then for a map of complexes $a^\bullet : M^\bullet \to K^\bullet$, the following are equivalent*
   (a) *$a^\bullet$ induces the zero map $\mathrm{Ext}^1_R(K^\bullet, N) \to \mathrm{Ext}^1_R(M^\bullet, N)$ for all R-modules N, and*

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(b) *there is a map* $h^0 : M^0 \to K^{-1}$ *such that* $a^{-1} + h^0 \circ d_K^{-1} = 0$.

(3) *Assume* $K^i = 0$ *for* $i \leq -3$. *Let* $\alpha \in \mathrm{Hom}_{D(R)}(M^\bullet, K^\bullet)$. *If the composition of* α *with* $K^\bullet \to K^{-2}[2]$ *comes from an* R*-module map* $a : M^{-2} \to K^{-2}$ *with* $a \circ d_M^{-3} = 0$, *then* α *can be represented by a map of complexes* $a^\bullet : M^\bullet \to K^\bullet$ *with* $a^{-2} = a$.

(4) *In (2) for any second map of complexes* $(a')^\bullet : M^\bullet \to K^\bullet$ *representing* α *with* $a = (a')^{-2}$ *there exist* $h^i : M^i \to K^{i-1}$ *for* $i = 0, -1$ *such that*

$$h^{-1} \circ d_M^{-2} = 0, \quad (a')^{-1} = a^{-1} + d_K^{-2} \circ h^{-1} + h^0 \circ d_M^{-1}, \quad (a')^0 = a^0 + d_K^{-1} \circ h^0$$

Proof. Set $F^0 = M^0$. Choose a free R-module F^{-1} and a surjection $F^{-1} \to M^{-1}$. Choose a free R-module F^{-2} and a surjection $F^{-2} \to M^{-2} \times_{M^{-1}} F^{-1}$. Continuing in this way we obtain a quasi-isomorphism $p^\bullet : F^\bullet \to M^\bullet$ which is termwise surjective and with F^i projective for all i.

Proof of (1). By Derived Categories, Lemma 064B we have

$$\mathrm{Hom}_{D(R)}(M^\bullet, K^\bullet) = \mathrm{Hom}_{K(R)}(F^\bullet, K^\bullet)$$

If $K^i = 0$ for $i \leq -2$, then any morphism of complexes $F^\bullet \to K^\bullet$ factors through $p^\bullet$. Similarly, any homotopy $\{h^i : F^i \to K^{i-1}\}$ factors through $p^\bullet$. Thus (1) holds.

Proof of (2). If (2)(b) holds, then $a^\bullet$ is homotopic to a map of complexes $(a')^\bullet : M^\bullet \to K^\bullet$ which is zero in degree -1. On the other hand, let $N \to I^\bullet$ be an injective resolution. We have

$$\mathrm{Ext}^1_R(K^\bullet, N) = \mathrm{Hom}_{D(R)}(K^\bullet, I^\bullet[1]) = \mathrm{Hom}_{K(R)}(K^\bullet, I^\bullet[1])$$

by Derived Categories, Lemma 05TG. Let $b^\bullet : K^\bullet \to I^\bullet[1]$ be a map of complexes. Since $K^1 = 0$ the map $b^0 : K^0 \to I^1$ maps into the kernel of $I^1 \to I^2$ which is the image of $I^0 \to I^1$. Since K^0 is projective we can lift b^0 to a map $h : K^0 \to I^0$. Thus we see that $b^\bullet$ is homotopic to a map of complexes $(b')^\bullet$ with $(b')^0 = 0$. Since $K^i = 0$ for $i \notin [-1, 0]$ it follows that $(b')^\bullet \circ (a')^\bullet = 0$ as a map of complexes. Hence the map $\mathrm{Ext}^1_R(K^\bullet, N) \to \mathrm{Ext}^1_R(M^\bullet, N)$ is zero. In this way we see that (2)(b) implies (2)(a). Conversely, assume (2)(a). We see that the canonical element in $\mathrm{Ext}^1_R(K^\bullet, K^{-1})$ maps to zero in $\mathrm{Ext}^1_R(M^\bullet, K^{-1})$. Using (1) we see immediately that we get a map h^0 as in (2)(b).

Proof of (3). Choose $b^\bullet : F^\bullet \to K^\bullet$ representing α. The composition of α with $K^\bullet \to K^{-2}[2]$ is represented by $b^{-2} : F^{-2} \to K^{-2}$. As this is homotopic to $a \circ p^{-2} : F^{-2} \to M^{-2} \to K^{-2}$, there is a map $h : F^{-1} \to K^{-2}$ such that $b^{-2} = a \circ p^{-2} + h \circ d_F^{-2}$. Adjusting $b^\bullet$ by h viewed as a homotopy from $F^\bullet$ to $K^\bullet$, we find that $b^{-2} = a \circ p^{-2}$. Hence b^{-2} factors through p^{-2}. Since $F^0 = M^0$ the kernel of p^{-2} surjects onto the kernel of p^{-1} (for example because the kernel of $p^\bullet$ is an acyclic complex or by a diagram chase). Hence b^{-1} necessarily factors through p^{-1} as well and we see that (3) holds for these factorizations and $a^0 = b^0$.

Proof of (4) is omitted. Hint: There is a homotopy between $a^\bullet \circ p^\bullet$ and $(a')^\bullet \circ p^\bullet$ and we argue as before that this homotopy factors through $p^\bullet$. □